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FACULTY OF CHEMICAL ENGINEERING



MECHANICAL PROCESSING

Laboratory Of Unit Operations (CH3015)

CONVECTION DRYING

CLASS CC03 — GROUP 9 — SEMESTER 252

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1 OBJECTIVES

Investigation of the convection drying for:

- Producing the drying and drying rate curves.
- Determining the data of drying such as constant drying rate, critical moisture content, equilibrium moisture content, constant-rate period and falling-rate period.
- Evaluating error of drying process.

2 THEORY

2.1 Definition

Convection drying is a process to remove water from the solid material by thermal vaporization. Wherein, the heat and moisture transfers are conducted by convection.

2.2 Characteristic of drying process

Drying, a complex process owns the irreversible and unstable properties. There are four concurrent processes occur in drying, which is heat transfer for solid material, diffusion of moisture inside the material, phase change and vaporization of the moisture into the ambient air.

2.3 Determination of drying rate based on heat balance of drying process

Heat duty of drying agent flow applies to a dryer in a period of $d\tau$:

$$dQ = \alpha \cdot F \cdot (t - \theta) d\tau \quad (1)$$

This heat duty is used for:

- Heating the solid material:

$$(G_0 C_0 + G_a C_a) d\theta \quad (2)$$

- Vaporizing the moisture and overheating the vapor:

$$[r + C_h(t - t_h)] dG_a \quad (3)$$

Where:

- α : Heat-transfer coefficient drying agent into the dried material, W/m^2K
- F : surface area of material, m^2
- t, θ, t_h : temperature of drying agent, material and saturated vapor, respectively, $^{\circ}C$
- G_0, C_0 : weight (kg) and specific heat ($J/Kg^{\circ}C$) of dried material, respectively
- G_a, C_a : Weight (Kg) and specific heat ($J/Kg^{\circ}C$) of moisture, respectively
- r : Latent heat of moisture, J/Kg
- C_h : Specific heat of moisture vapor, $J/Kg^{\circ}C$

Moisture content vaporized in a period of $d\tau$:

$$dG_a = d(G_0 U) = G_0 dU \quad (4)$$

Where, U : moisture content of material (kg moisture/ kg dry material).

From equation (1), (2), (3) and (4), a heat balance is established as following:

$$\alpha \cdot F \cdot (t - \theta) d\tau = (G_0 C_0 + G_a C_a) d\theta + G_0 [r + C_h(t - t_h)] dU \quad (5)$$

From (5), the drying rate inferred from heat balance is:

$$\frac{dU}{d\tau} = \frac{\alpha F(t - \theta) - (G_0 C_0 + G_a C_a) \frac{d\theta}{d\tau}}{G_0 [r + C_h(t - t_h)]} \quad (6)$$

2.4 The basic equation of drying kinetics

According to the equation of moisture transfer from the material into drying agent medium:

$$\frac{dU}{d\tau} = \frac{k_p F}{G_0} (p_m - p) \quad (7)$$

Where:

- k_p : moisture-transfer coefficient in gas phase, $kg/(m^2 \cdot h \cdot \Delta p)$.
- p_m, p : pressure of moisture on the material's surface and in the gas phase, respectively, $mmHg$ (at).

As the moisture is not overheated (*i.e.* $t = t_h$), the equation (5) is altered as follows:

$$\left(C_0 + C_a \frac{G_a}{G_0} \right) G_0 \frac{d\theta}{d\tau} + r G_0 \frac{dU}{d\tau} = \frac{dQ}{F d\tau} F = q F \quad (8)$$

With $\frac{G_a}{G_0} = U$; $\frac{G_0}{V_0} = \rho_0$; $C_0 + C_a U = C$ and $\frac{V_0}{F} = R_0$.

Wherein:

- ρ_0 : specific weight of dry material, kg/m^3
- V_0 : volume of dry material, m^3
- C : specific heat of moist material, $J/(kg \cdot ^\circ C)$
- R_0 : equivalent radius of material, m

If the heat duty for overheating the moisture vapor is ignored, we get:

$$q = \rho_0 R_0 r \frac{dU}{d\tau} + C \rho_0 R_0 \frac{d\theta}{d\tau} = \left[1 + \left(\frac{C}{r} \right) \frac{d\theta}{dU} \right] \left(\rho_0 R_0 r \frac{dU}{d\tau} \right) = (1 + Rb) (\rho_0 R_0 r) \frac{dU}{d\tau} \quad (9)$$

With $Rb = 1 + \left(\frac{C}{r} \right) \frac{d\theta}{dU}$: Rebinder number, representative for drying kinetics. Expression (10) is the basic equation of drying kinetics, expressing the change of moisture in a material with drying time.

2.5 Heat duty applies to material in falling-rate period (q_2)

In falling rate period of the drying process, the obtained drying rate curve is straight. Therefore, the drying rate in this period is expressed as:

$$-\frac{dU}{d\tau} = K(U - U^*) \quad (10)$$

Where K is the drying coefficient (slope drying rate curve), depended on drying rate and characteristics of moist material, $1/s$:

$$K = \frac{N}{U_{th} - U^*} = \chi N \quad (11)$$

Where:

- $\chi = \frac{1}{U_{th} - U^*}$: relative drying coefficient, depended on moist material
- U_{th} : critical moist content
- U^* : equilibrium moisture content
- N : constant drying rate, $kg \text{ moisture} / (kg \text{ dry material} \cdot s)$

Integral of equation (10) we get:

$$\frac{U - U^*}{U_{th} - U^*} = \exp(-\chi N \tau) \quad (12)$$

Alternatively, logarithmic solving of equation (10):

$$\lg(U - U^*) = \lg(U_{th} - U^*) - \frac{1}{2.3} \chi N \tau \quad (13)$$

If the drying coefficient is determined via experiments and can be approximately calculated as following:

$$\chi \approx \frac{1.8}{U_0} \quad (14)$$

With U_0 : the initial moisture content of material.

Therefore:

$$U_{th} = \frac{1}{\chi} + U^* = \frac{U_0}{1.8} + U^* \quad (15)$$

Replace equation (10) with equation (11) and (14), then:

$$-\frac{dU}{d\tau} = 1.8N \left(\frac{U - U^*}{U_0} \right) \quad (16)$$

Combining equation (16) and (10), we get:

$$q_2 = (1 + Rb)(\rho_0 R_0 r) 1.8N \left(\frac{U - U^*}{U_0} \right) \quad (17)$$

2.6 Heat duty applies to material in constant-rate period (q_1)

In constant rate period, the total heat duty is equal to the heat of vaporization and the material temperature is constant:

$$q_1 = \rho_0 R_0 r \frac{dU}{d\tau} = \rho_0 R_0 r N \quad (18)$$

2.7 Heat flux of heat-transfer density ($q(x)$)

$$q(x) = \frac{q_2}{q_1} = 1.8 \frac{U - U^*}{U_0} (1 + Rb) \quad (19)$$

2.8 Drying and drying rate curves

2.8.1 Drying curve

It expresses the variation of moisture content (U) with drying time (τ):

$$U = f(\tau) \quad (20)$$

The shape of drying curve:

- Depended on various factors, such as moisture-material bonding, shape, dimension, the structure of the material; drying method and drying condition.
- Drying curve is a function of the drying process. Therefore, the shape of drying curve is the same despite different drying methods and or drying conditions.

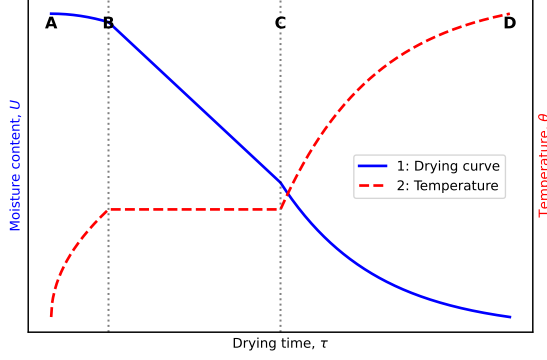


Figure 1: Drying curve

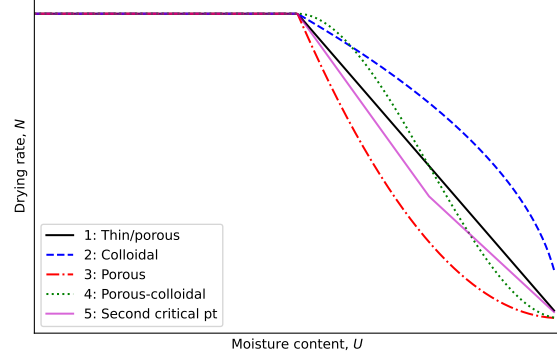


Figure 2: Drying rate curve

Figure Legend:

AB: Heating of material

BC: constant rate drying

CD: Falling rate drying

1: drying curve

2: temperature pattern of material

1: thin and porous material: paper, paperback...

2: colloidal material

3: porous material

4: porous-colloidal material: has inflection point

5: material has break point (second critical point)

2.9 The drying stages

2.9.1 Warming-up period (AB)

The warming-up period occurs rapidly, just in very short time. The total heat duty of drying agent was used for heating the material from initial temperature (θ_0) to the wet-bulb temperature (t_u) (Fig 1). In this stage, the moisture content is vaporized negligibly, and the reduction of humidity of material is very tiny. The drying rate drastically increases to a maximum value (N). In fact, the stage is ignored in drying calculation.

2.9.2 Constant rate period (BC)

In this period, the diffusion rate of moisture from inside to the surface of material is higher than that from the surface to the heating medium. Therefore, the surface of material is covered by the moisture. Total heat duty is used for vaporizing moisture on the surface of the material (free moisture), and the vaporized surface stays constant. Therefore, the temperature on the surface of the material and drying rate do not alter when the moisture content rapidly decreases. Drying time of this stage (constant-rate time, τ_1) is determined as follow:

$$-\frac{dU}{d\tau} = N_1 = \text{const} \quad (21)$$

Integral of equation (23), we get:

$$\tau_1 = \frac{U_0 - U_{th}}{N_1} \quad (22)$$

Where U_{th} : critical moisture content, which is the moisture content at the end of the constant-rate drying period.

2.9.3 Falling-rate drying period (CD)

In this stage, the moisture on the surface of material is completely vaporized, so the vaporizing surface is restricted to inside the material. The diffusion rate of moisture in the material is slow and hence, gradually reduces the drying rate. The temperature of the material rises from the wet-bulb temperature (t_u) to the dry-bulb temperature (t). At this time, there are three regions in the material: moist region, vaporizing region, and dry region.

If the drying rate curve of this stage is straight or converting to a straight line ($-N_2 = ax + b$), we can integrate for determination of the falling period of drying process (τ_2):

$$\tau_2 = \frac{U_{th} - U^*}{N_1} \ln \frac{U_{th} - U^*}{U_2 - U^*} \quad (23)$$

Where U^* : equilibrium moisture content, which is the moisture content at the end of the falling-rate drying period.

2.10 Drying time

Drying time can be determined via summarizing the drying time of three stages, including warming-up (τ_0), constant-rate (τ_1) and falling rate period (τ_2). The warming-up stage can be ignored, the equation expressing the drying time is:

$$\tau = \tau_1 + \tau_2 = \frac{U_0 - U_{th}}{N} + \frac{2.3}{N}(U_{th} - U^*) \lg \left[\frac{U_{th} - U^*}{U_2 - U^*} \right] \quad (24)$$

Where U_2 : the moisture content at the end of the drying process, corresponding to (τ_2). $U_2 > U^*$, usually using $U_2 = U^* + (2 \div 3)\%$.

3 EXPERIMENTAL APPARATUS AND MATERIAL

3.1 Experimental apparatus

The drying apparatus consists of:

- **Calorifer:** Has two dry resistors with a capacity of 10KW. Temperature is stabilized by an automatic switch.
- **Sucking blower:** Sucking speed of 0.85m/s for loading moist air (drying agent) and blowing through the calorifer to raise the temperature.

3.2 Material

Filter paper is used as the material for drying.

4 EXPERIMENTAL METHOD

4.1 Content

Conducting the experimental drying at three different temperatures of calorifer: $50^\circ C$, $60^\circ C$, and $70^\circ C$.

4.2 Experiments

4.2.1 Observation of dryer

Observe the drying apparatus and compare with the diagram before practice.

4.2.2 Preparation

- Weighing the initial dry filter paper (G_0) using the mass balance of the dryer.
- Sufficiently humidifying the filter paper, then determining the weight (G).
- Preparing and setting zero for the timer.

4.2.3 Dryer checking

- Close the door of the drying oven.
- Check the valve of air inlet and outlet doors.
- Add water into the water gourd of the wet-bulb thermometer.

4.2.4 Start of the dryer

- **Blower:** Switch on for loading humid air and feeding to the calorifer.
- **Calorifer:** Switch on. Turn on the first resistor for 50°C , and the second for higher temperatures.

4.3 Collecting data in an experiment

4.3.1 The data need collecting

The weight of filter paper before, after and during drying process, dry and wet-bulb temperature, drying time.

4.3.2 Reporting the weight by mass balance

- The weight (g): reporting the small value on the mass balance
- Temperature ($^{\circ}\text{C}$): reporting the value on the temperature indicator

4.4 Accomplishing

1. Turn off all dry resistors of the calorifer.
2. Switch off the blower after shutting down the calorifer.
3. Remove filter paper from the drying oven.
4. Switch off the circuit breaker (CP) of the dryer.

5 REPORT

5.1 Experiment data

Table 1: Raw data for Drying Temperatures 50°C , 60°C , and 70°C

Drying Regime 50°C				Drying Regime 60°C				Drying Regime 70°C			
τ (min)	G (g)	T_u	T_k	τ (min)	G (g)	T_u	T_k	τ (min)	G (g)	T_u	T_k
0	65	28	46	0	70	29	59	0	65	30	69
5	65	28	53	3	67.5	29	56	2	65	31	60
10	65	27	52	6	67.5	29	62	4	65	30	62
15	65	27	45	9	65	29	64	6	62.5	30	64
20	62.5	26	57	12	65	30	65	8	62.5	30	66
25	62.5	27	48	15	65	30	66	10	62.5	30	67
30	62.5	27	55					12	62.5	30	72

5.2 Calculated data

Table 2: Calculated data for 50°C

$\tau(min)$	$G(kg)$	$U(\%)$	$\Delta U(\%)$	$N(\%/h)$	$t_w(^{\circ}C)$	$t_k(^{\circ}C)$	$P_m(mmHg)$	$P(mmHg)$	ϵ
0.00	0.0650	4.00	0.00	0.00	28	46	28.27	19.27	18
0.08	0.0650	4.00	0.00	0.00	28	53	28.27	15.77	25
0.17	0.0650	4.00	0.00	0.00	27	52	26.66	14.16	25
0.25	0.0650	4.00	0.00	0.00	27	45	26.66	17.66	18
0.33	0.0625	0.00	4.00	48.00	26	57	25.14	9.64	31
0.42	0.0625	0.00	0.00	0.00	27	48	26.66	16.16	21
0.50	0.0625	0.00	0.00	0.00	27	55	26.66	12.66	28

Table 3: Calculated data for 60°C

$\tau(min)$	$G(kg)$	$U(\%)$	$\Delta U(\%)$	$N(\%/h)$	$t_w(^{\circ}C)$	$t_k(^{\circ}C)$	$P_m(mmHg)$	$P(mmHg)$	ϵ
0.00	0.0700	7.69	0.00	0.00	29	59	29.96	14.96	30
0.05	0.0675	3.85	3.85	76.92	29	56	29.96	16.46	27
0.10	0.0675	3.85	0.00	0.00	29	62	29.96	13.46	33
0.15	0.0650	0.00	3.85	76.92	29	64	29.96	12.46	35
0.20	0.0650	0.00	0.00	0.00	30	65	31.74	14.24	35
0.25	0.0650	0.00	0.00	0.00	30	66	31.74	13.74	36

Table 4: Calculated data for 70°C

$U(\%)$	U_{th}	$P_m(mmHg)$	$P(mmHg)$	J_m	α	f	N	N_{tb}	c	K	τ_1	τ_2
0.00	0.0650	4.00	0.00	0.00	30	69	31.74	12.24	39			
0.03	0.0650	4.00	0.00	0.00	31	60	33.61	19.11	29			
0.07	0.0650	4.00	0.00	0.00	30	62	31.74	15.74	32			
0.10	0.0625	0.00	4.00	120.00	30	64	31.74	14.74	34			
0.13	0.0625	0.00	0.00	0.00	30	66	31.74	13.74	36			
0.17	0.0625	0.00	0.00	0.00	30	67	31.74	13.24	37			
0.20	0.0625	0.00	0.00	0.00	30	72	31.74	10.74	42			

5.3 Plots

5.3.1 The drying curve

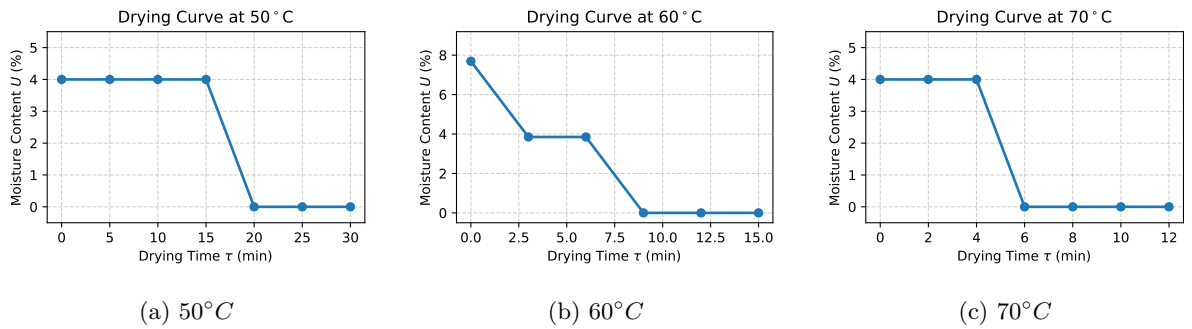


Figure 3: Experimental Drying Curves.

5.3.2 The drying rate curve

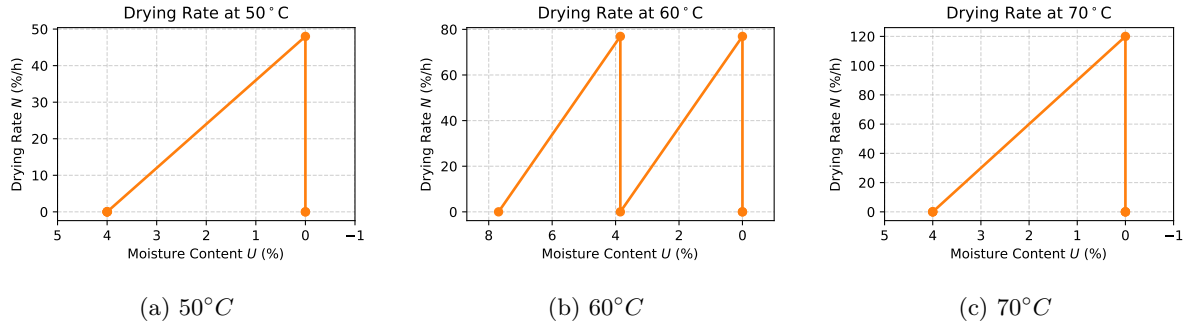


Figure 4: Experimental Drying Rate Curves.

5.4 Evaluation of results

5.4.1 Calculated results in theory

Table 5: Theoretical calculations for 50°C

$U(\%)$	U_{th}	$P_m(mmHg)$	$P(mmHg)$	J_m	α	f	N	N_{tb}	c	K	τ_1	τ_2
217.50	-	46	37.86	0.31	-	-	-	276.12	-	-	-	-
187.50	-	47	42.30	0.18	-	-	-	159.43	-	-	-	-
150.00	-	48	43.02	0.19	-	-	-	168.93	-	-	-	-
100.00	-	47	42.30	0.18	-	-	-	159.43	-	-	-	-
62.50	-	47	40.86	0.23	-	-	-	208.27	-	-	-	-
31.25	-	47	41.82	0.20	-	-	-	175.71	-	-	-	-
12.50	-	46	39.30	0.25	-	-	-	227.27	-	-	-	-
6.25	-	47	40.38	0.25	-	-	-	224.56	-	-	-	-
0.00	-	46	38.82	0.27	-	-	-	243.55	-	-	-	-
0.00	-	47	39.90	0.27	-	-	-	240.84	-	-	-	-
0.00	-	46	39.30	0.25	-	-	-	227.27	-	-	-	-
Summary	120.83	-	-	-	0.04	9.00	210.13	-	0.46	2.13	1.74	0.01

Table 6: Theoretical calculations for 60°C

$U(\%)$	U_{th}	$P_m(mmHg)$	$P(mmHg)$	J_m	α	f	N	N_{tb}	c	K	τ_1	τ_2
217.50	-	68	60.79	0.29	-	-	-	259.56	-	-	-	-
175.00	-	68	60.32	0.31	-	-	-	276.48	-	-	-	-
131.25	-	65	56.83	0.33	-	-	-	294.12	-	-	-	-
87.50	-	65	57.30	0.31	-	-	-	277.20	-	-	-	-
56.25	-	65	57.30	0.31	-	-	-	277.20	-	-	-	-
31.25	-	65	57.78	0.29	-	-	-	259.92	-	-	-	-
6.25	-	65	57.78	0.29	-	-	-	259.92	-	-	-	-
0.00	-	65	57.78	0.29	-	-	-	259.92	-	-	-	-
0.00	-	65	58.25	0.27	-	-	-	243.00	-	-	-	-
0.00	-	65	57.78	0.29	-	-	-	259.92	-	-	-	-
Summary	120.83	-	-	-	0.04	9.00	266.72	-	0.36	2.21	1.67	0.01

Table 7: Theoretical calculations for 70°C

$U(\%)$	U_{th}	$P_m(mmHg)$	$P(mmHg)$	J_m	α	f	N	N_{tb}	c	K	τ_1	τ_2
217.50	-	70	61.15	0.33	-	-	-	300.20	-	-	-	-
175.00	-	67	58.91	0.30	-	-	-	274.42	-	-	-	-
143.75	-	67	57.50	0.36	-	-	-	322.25	-	-	-	-
112.50	-	72	64.51	0.28	-	-	-	254.07	-	-	-	-
87.50	-	67	58.44	0.32	-	-	-	290.36	-	-	-	-
62.50	-	70	60.22	0.37	-	-	-	331.75	-	-	-	-
31.25	-	67	58.44	0.32	-	-	-	290.36	-	-	-	-
18.75	-	70	61.15	0.33	-	-	-	300.20	-	-	-	-
0.00	-	70	62.56	0.28	-	-	-	252.37	-	-	-	-
0.00	-	67	57.50	0.36	-	-	-	322.25	-	-	-	-
0.00	-	75	69.44	0.21	-	-	-	188.60	-	-	-	-
Summary	120.83	-	-	-	0.04	9.00	284.26	-	0.34	2.35	1.57	0.01

5.4.2 The error

Table 8: Error evaluation for U_{th} and N

Drying Temp	$U_{th}(\%)$			$N(\%/h)$		
	Experiment	Theory	Error (%)	Experiment	Theory	Error (%)
50°C	73.030	120.830	-39.560	484	210.130	130.334
60°C	25.510	120.830	-78.888	820	266.720	207.439
70°C	14.990	120.830	-87.594	901.82	284.260	217.252

Table 9: Error evaluation for χ and K

Drying Temp	Relative drying coefficient χ			Drying Coefficient K (1/h)		
	Experiment	Theory	Error (%)	Experiment	Theory	Error (%)
50°C	0.298	0.460	-35.110	0.482	2.130	-77.386
60°C	0.234	0.360	-34.963	0.067	1.670	-96.013
70°C	0.225	0.340	-33.954	0.027	1.570	-98.297

Table 10: Experimental vs Theoretical summary for χ and N

Drying Temp	Relative density χ			Constant Drying Rate $N(\%/h)$		
	Experiment	Theory	Error (%)	Experiment	Theory	Error (%)
50°C	0.01	0.010	0.000	6.620	1.740	280.460
60°C	0.04	0.010	292.003	32.144	2.210	1354.491
70°C	0.07	0.010	567.111	60.161	2.350	2460.061

6 DISCUSSION

6.1 Shape of drying curves compared to theory

- **Warming-up:** Not clearly visible in experimental curves because it occurs very fast (ignored recordings usually start at 2-3 mins).
- **Constant vs. Falling:** Visible, though transition points (critical moisture U_{th}) are sensitive to sampling frequency.
- **Temperature Effect:** Consistent with theory; higher temperatures yield higher drying rates (lower curves for lower temperatures).

6.2 Analysis of results

- **Constant Rate N :** Matches theory conceptually; higher T yields higher N .
- **Relative drying coefficient χ :** Theoretically only dependent on material, but experimental values vary because they rely on calculated U_{th} and U^* .
- **Drying coefficient K :** Suitable trend; K increases with temperature as N increases.
- **Drying time τ :** τ_1 and τ_2 generally decrease with increasing temperature, though experimental errors in N affect τ_1 suitability.

6.3 Error evaluation

- **Critical moisture U_{th} :** Large errors (39% ÷ 87%) due to plotting approximations.
- **Constant rate N :** Very large errors (130% ÷ 217%). Causes include:
 - Assuming constant α_p (empirical formula $\alpha_p = 0.0229 + 0.0174v_k$).
 - Inaccuracies in determining P_m and P from graphs.
 - Manual plotting of the drying curve differentials.
- **Time τ :** Negative errors in τ_1 (Exp < Theory) because experimental N was significantly larger than theoretical predictions.

6.4 Methods for remediation

- **Experiment:** Ensure materials are 100% dry for G_c , use electronic sensors for precise recording.
- **Calculation:** Use formulas instead of manual graph lookups, apply minimum mean square error for plotting.

7 CALCULATION

7.1 For Table 1 & 2 (Treated Data)

$$U(\%) = \frac{G - G_c}{G_c} \times 100 \quad (25)$$

$$N(\%/h) = \frac{\Delta U}{\Delta \tau} \quad (26)$$

$$\epsilon = t_k - t_w \quad (27)$$

7.2 For Table 4 (Results from Plots)

- U_{th} : Intersection of drying and drying rate curves.
- U^* : Equilibrium moisture content (assumed 0 for experimental filter paper).
- U_2 : $U^* + (2 \div 3)\%$.
- $\chi = \frac{1}{U_{th} - U^*}$.
- $K = \chi N$.

7.3 For Table 5 (Theory vs. Experiment)

$$U_{th_theory} = \frac{U_0}{1.8} + U^* \quad (28)$$

$$\alpha_p = 0.0229 + 0.0174v_k \quad [kg/(m^2 \cdot h \cdot mmHg)] \quad (29)$$

$$N_{theory} = 100 \cdot \alpha_p \cdot (P_m - P) \cdot \frac{F}{G_c} \quad [%/h] \quad (30)$$

Where $F = 0.36m^2$ (evaporation area).

7.4 Error Calculation

$$\text{Error (\%)} = \frac{\text{Experiment} - \text{Theory}}{\text{Theory}} \times 100 \quad (31)$$

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